

SmagerUp Core (SUC) - Updated System Documentation

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Overview

This update finalizes the SmagerUp Core (SUC) architecture, focusing on modularity, consistency, and scalability.

SUC now serves as the secure backend for SmagerUp Studio (SUS), managing module hosting, licensing, authentication, and API responses.

1. Key Enhancements

Unified Response Format

All controllers now use a centralized response structure via `ControllerExtensions` and `ApiResponse<T>`:

```
{
  "ok": true,
  "msg": "",
  "data": { ... }
}
```

Token-Based Authentication

`TokenService.cs` now generates secure JWT tokens using HMAC SHA256 with 24-hour expiry.

Claims:

- `NameIdentifier` (`AccountId`)
- `Name` (Account Name)
- `Jti` (Unique Token ID)

Modular Access Control

ModulesController handles both free and paid module logic:

- Free (LicenseTypeId = 1): Always accessible.
- Paid: Requires valid license or active AccountModule entry.

Dapper Data Layer

The old ORM has been replaced with Dapper for high-performance, lightweight database access.

Repositories:

- AccountRepository
- ModuleRepository
- LicenseTypeRepository
- ContentRepository
- DapperContext

DTOs and Models

Refined structure for modular and clean data representation.

DTOs:

- ModuleResponseDto
- ValidateRequestDto

Models:

- Account
- Module
- AccountModule
- LicenseType
- Content
- ContentGroup
- ApiResponse

2. Database Schema Alignment

- ModuleId and AccountModuleId use Guid (uniqueidentifier).
- LicenseType defines free and commercial licenses.
- Contents and ContentGroups now handle HTML/CSS/JS storage dynamically.

3. Code Structure Overview

Controllers/

- AuthController.cs
- ModulesController.cs
- LicenseTypesController.cs
- ControllerExtensions.cs
- SucController.cs

Data/

- AccountRepository.cs
- ModuleRepository.cs
- LicenseTypeRepository.cs
- ContentRepository.cs
- DapperContext.cs

DTOs/

- ModuleResponseDto.cs
- ValidateRequestDto.cs

Models/

- Account.cs
- Module.cs
- AccountModule.cs
- LicenseType.cs
- Content.cs
- ContentGroup.cs

- ApiResponse.cs

Services/

- TokenService.cs

4. Response Examples

Success

```
{  
  "ok": true,  
  "msg": "",  
  "data": {  
    "name": "grid",  
    "version": "1.0.0",  
    "license": "SFUL",  
    "price": 0.00,  
    "contents": []  
  }  
}
```

Fail

```
{  
  "ok": false,  
  "msg": "need to purchase or license expired",  
  "data": {  
    "name": "grid",  
    "version": "1.0.0",  
    "license": "SCL",  
    "price": 49.99,  
    "contents": []  
  }  
}
```

}

5. Summary

SmagerUp Core (SUC) now provides a fully modular, secure, and license-aware foundation.
Ready for seamless integration with SmagerUp Studio (SUS).